

Sentiment-to-Price Correlation Analysis Report

1. Executive Summary

This report analyzes the linear correlation between FinBERT-extracted news sentiment and next-day stock returns across 20 major NSE equities.

- Total Data Samples Analyzed: 2583 trading days
- Overall Pearson Correlation (\$r\$): 0.038
- Statistical Significance (\$p\$-value): 0.0538
- Overall Assessment: Weak / Moderate Correlation

2. Correlation Breakdown by NSE Stock

Symbol	Pearson \$r\$	\$p\$-value	Significant (p < 0.05)	Sample Size
WAAREEENER	0.2334	0.0094	Yes (p < 0.05)	123
WIPRO	0.1855	0.0400	Yes (p < 0.05)	123
AXISBANK	0.1098	0.2265	No	123
ITC	0.1026	0.2588	No	123
SUNPHARMA	0.0713	0.4331	No	123
LT	0.0634	0.4857	No	123
TCS	0.0630	0.4886	No	123
BHARTIARTL	0.0438	0.6305	No	123
HDFCBANK	0.0348	0.7023	No	123
BAJFINANCE	0.0301	0.7411	No	123
SBIN	0.0087	0.9235	No	123
HINDUNILVR	-0.0118	0.8967	No	123
INFY	-0.0180	0.8435	No	123
TITAN	-0.0189	0.8356	No	123
ASIANPAINT	-0.0212	0.8160	No	123
NTPC	-0.0263	0.7724	No	123
MARUTI	-0.0317	0.7282	No	123
ICICIBANK	-0.0392	0.6670	No	123
HCLTECH	-0.0405	0.6562	No	123
RELIANCE	-0.0993	0.2747	No	123
KOTAKBANK	-0.1211	0.1822	No	123

3. Key Observations & Findings

1. Stocks with strong positive sentiment-return alignment display predictive signals suitable for reinforcement learning trading state representation.
2. News volume and sentiment variance serve as important indicators of price volatility on subsequent trading days.
3. Lagged features (1-day and 2-day prior sentiment) often carry stronger predictive signal than same-day sentiment due to news dissemination delays.